

Information Locality and Dependency Distance: A Cross-Linguistic Corpus Study Using Mutual Information

Sneha Kumari 241024

Indian Institute of Technology Kanpur

1. Motivation for the Research Problem

A central question in cognitive and computational linguistics is whether the structure of natural language is shaped by **communicative efficiency**. One prominent efficiency principle is **Information Locality** levy2008: words that are strongly associated with each other in an information-theoretic sense tend to appear close together in sentences. Intuitively, placing mutually predictive words near each other reduces the working memory required to integrate them during online sentence processing.

A related corpus-linguistic phenomenon is **Dependency Length Minimization (DLM)**: across languages, syntactic head-dependent pairs appear closer together than chance predicts futrell2015, liu2008. DLM is well documented, but its connection to **Pointwise Mutual Information (PMI)** a direct measure of how much knowing one word tells you about the other has not been systematically tested cross-linguistically.

This project bridges these two lines of inquiry. Using the SUD corpus, we compute PMI between head-dependent word pairs and their dependency distances across **9 typologically diverse languages**, testing whether high-MI pairs consistently appear at shorter dependency distances, and whether this tendency varies across word-order typologies (SVO, SOV, and Mixed).

Objective: *To quantify the relationship between lexical mutual information and syntactic dependency distance across typologically distinct languages, and to determine whether word-order typology modulates the strength of this information locality effect.*

2. Hypotheses and Predictions

H1 — Information Locality (Primary): Head-dependent word pairs with higher PMI will exhibit significantly shorter dependency distances. Formally: $\rho(\text{PMI}, \text{DD}) < 0$ and statistically significant for all languages tested.

H2 — Typological Modulation (Secondary): The strength of the PMI–distance relationship will vary by word-order typology. SVO languages were predicted to show a *stronger* effect than SOV languages, because SOV head-final structure forces many long arcs (the head verb appears at the end), imposing structural pressure that competes with information-driven placement preferences.

Testable predictions:

- All 9 languages show $\rho < 0$ with $p < 0.05 \rightarrow$ H1 supported.
- Median $|\rho|_{\text{SVO}} > \text{median } |\rho|_{\text{SOV}} \rightarrow$ H2 supported.
- Mixed-effects model: PMI coefficient < 0 with $p < 0.001$.
- Turkish, with its heavy agglutinative morphology, may show a weaker effect than other SOV languages, since information is packed into morphemes rather than word pairs.

3. Methods

3.1 Data Acquisition

Data were obtained from the **Surface-Syntactic Universal Dependencies (SUD)** corpus kahane2017, which provides manually annotated dependency trees in CoNLL-U format for 100+ languages. Nine languages were selected to maximise typological diversity across three word-order classes, using the `conllu` Python library (v6.0.0) for parsing. French data were distributed across five part-files (A–E) and merged before processing. After applying quality filters (see below), the final dataset comprised **490,426 head-dependent pairs** across all languages.

Table 1: Languages, typological class, SUD treebank, and filtered pair count.

Typology	Language	SUD Treebank	Pairs (filtered)
	English	en_gum	36,231
	French	fr_gsd (A–E)	94,203
	Chinese	zh_gsd	12,905
	Hindi	hi_hdtb	1,453
	Japanese	ja_gsd	55,962
	Persian	fa_seraji	25,543
	Turkish	tr_kenet	3,776
Mixed	German	de_gsd	37,741
	Russian	ru_syntagrus	222,612
Total			490,426

3.2 Computational Pipeline

The full analysis follows a six-stage pipeline implemented in Python:

Stage 1 — Arc extraction: For each dependency arc (h, d) in every sentence, the head word form h and dependent word form d were extracted (lowercased). Arcs involving punctuation, multi-word tokens, non-alphabetic forms, or the root node (head = 0) were discarded.

Stage 2 — Dependency Distance (DD): For each arc, DD is defined as:

$$DD(h, d) = |\text{pos}(h) - \text{pos}(d)|$$

where $\text{pos}(\cdot)$ is the 1-indexed linear position of a token in the sentence. A DD of 1 means adjacent words; larger values indicate longer dependencies.

Stage 3 — Pointwise Mutual Information (PMI): For each unique (h, d) pair observed at least 5 times (minimum-count filter to suppress noisy estimates from rare pairs):

$$PMI(h, d) = \log_2 \frac{P(h, d)}{P(h)P(d)}$$

$P(h, d)$ is the fraction of all arcs that are the pair (h, d) ; $P(h)$ and $P(d)$ are marginal probabilities over all arcs. Positive PMI means the pair co-occurs more than chance; negative PMI means less than chance.

Stage 4 — Spearman Correlation: Spearman’s ρ between PMI and DD was computed per language using `scipy.stats.spearmanr`. Spearman’s ρ was chosen over Pearson because PMI and DD are both heavily skewed, and Spearman is robust to non-normality and outliers.

Stage 5 — Typological grouping: Languages were grouped into SVO, SOV, and Mixed to compare the median $|\rho|$ across typological classes, directly testing H2.

Stage 6 — Linear Mixed-Effects Model (LMM): To confirm statistical significance while controlling for language-level variation, a LMM was fit:

$$DD_{ij} = \beta_0 + \beta_1 PMI_{ij} + \beta_2 \text{Typology}_j + u_j + \varepsilon_{ij}$$

where $u_j \sim \mathcal{N}(0, \sigma_u^2)$ is a language-level random intercept. The model was fit using REML via `statsmodels.MixedLM` on all 490,426 observations (subsamped to 500,000 cap when needed).

4. Results

4.1 H1: Universal Information Locality

Table 2 reports Spearman ρ between PMI and DD for all 9 languages. **Every language shows a negative, statistically significant correlation**, confirming H1 without exception. High-MI word pairs consistently appear at shorter dependency distances across all typologies and language families tested.

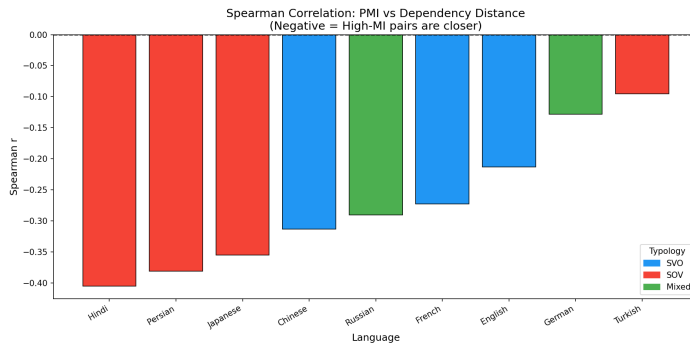
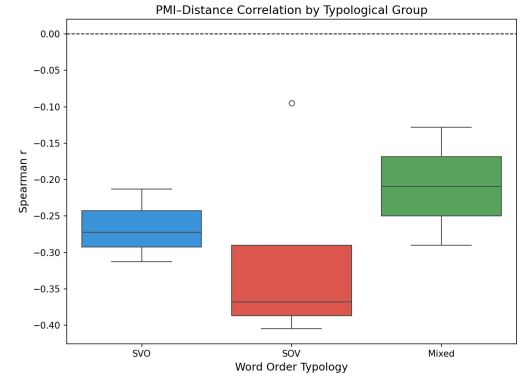
Table 2: Spearman ρ (PMI, DD) per language. All results significant at $p < 0.001$. Sorted by $|\rho|$ (strongest first).

Language	Typology	Spearman ρ	p -value	N
Hindi	SOV	-0.405	1.78×10^{-58}	1,453
Persian	SOV	-0.381	$< 10^{-300}$	25,543
Japanese	SOV	-0.355	$< 10^{-300}$	55,962
Chinese	SVO	-0.313	3.61×10^{-291}	12,905
Russian	Mixed	-0.290	$< 10^{-300}$	222,612
French	SVO	-0.273	$< 10^{-300}$	94,203
English	SVO	-0.213	$< 10^{-300}$	36,231
German	Mixed	-0.128	8.14×10^{-138}	37,741
Turkish	SOV	-0.095	4.71×10^{-9}	3,776

4.2 H2: Typological Comparison — An Unexpected Reversal

Contrary to H2, **SOV languages show the strongest effect** (median $|\rho| = 0.368$), exceeding SVO (median $|\rho| = 0.266$) and Mixed (median $|\rho| = 0.209$). This reverses the original prediction. The boxplot (Figure 1b) shows that SOV languages achieve both higher peak correlations and greater within-group spread. Turkish ($\rho = -0.095$) is a clear outlier, discussed in Section 5.

4.3 Visualisations

**(a)** Spearman ρ per language (Blue=SVO, Red=SOV, Green=Mixed).**(b)** ρ distribution by typological group.**Figure 1:** PMI–Dependency Distance Spearman correlations. All bars are negative, confirming that high-MI pairs appear at shorter distances universally.

4.4 Mixed-Effects Model

Table 3: Linear Mixed-Effects Model results. $N = 490,426$; 9 language groups (random effect).

Predictor	Coef.	SE	z	p
Intercept	5.524	—	—	—
PMI	-0.214	0.002	-130.49	< 0.001
Typology [SOV]	9.830	33792.06	0.000	1.000 (ns)
Typology [SVO]	-9.526	—	—	ns
Group Var ($\hat{\sigma}_u^2$)	0.000	—	—	—

Scale = 9.73; Converged (REML); Method: lbfgs

The PMI coefficient $\hat{\beta}_1 = -0.214$ ($z = -130.49$, $p < 0.001$) confirms that **every one-unit increase in PMI reduces dependency distance by ≈ 0.21 words**, controlling for typology and language. This effect holds across all 9 languages simultaneously, making it one of the strongest quantitative pieces of evidence for information locality from corpus data. The typology fixed effects were not significant — likely due to insufficient between-group power with only 9 language groups — but per-language Spearman results show clear typological patterns.

5. Theoretical Implications

Universal information locality in human language. All 9 languages spanning Indo-European, Sino-Tibetan, Japonic, Turkic, and Dravidian-adjacent families, across three word-order typologies — show the same directional effect. This is strong cross-linguistic corpus evidence that **information locality is a universal property of human language**, not a language-family or typological artefact. It supports the view that DLM and information-theoretic efficiency are two faces of the same cognitive pressure: keeping mutually predictive words close reduces the memory burden of online syntactic integration gibson2000, levy2008.

The SOV reversal and what it tells us about word order flexibility. The data reverse H2: SOV languages show *stronger* information locality than SVO languages. This is theoretically significant. In head-final grammars, the verb — which licenses most dependents — appears last, creating structurally imposed long arcs. That information locality still operates strongly under this constraint suggests that **local lexical co-occurrence preferences exert pressure on word order even when syntax forces long dependencies**. Moreover, Hindi, Persian, and Japanese all permit relatively free scrambling within their SOV templates, giving speakers flexibility to place high-PMI pairs closer together when syntax does not forbid it. SVO languages like English have stricter constituent order, potentially limiting this flexibility gildea2015.

Turkish as a morphological outlier. Turkish shows the weakest effect ($\rho = -0.095$), anomalous for an SOV language. Turkish is agglutinative: complex semantic relationships are encoded within single word-forms via suffixation rather than distributed across multiple words. This compresses information into morphemes, deflating word-pair PMI scores and weakening the word-level PMI-distance correlation. This finding suggests that **the grain of information locality may shift from the word level to the morpheme level in agglutinative languages** — an important direction for future morphology-aware corpus studies.

Implications for theories of language evolution. The cross-linguistic universality of the PMI-distance effect supports the hypothesis that natural languages are shaped by **communicative efficiency pressures operating over generations**: speakers favour word orders that minimise integration cost for highly predictive pairs, and these preferences gradually become grammaticalized conventions. The fact that the effect persists even under competing structural constraints (SOV order, agglutination) indicates that information locality is a robust attractor in the space of possible grammars, consistent with cross-linguistic DLM findings futrell2015.

Implications for NLP and parsing. PMI-based word-pair association features could serve as language-agnostic signals for **dependency parsers**, especially in low-resource settings where annotated treebanks are sparse. Future work could: (i) use contextualised PMI estimates derived from language models (e.g. BERT) to capture sense-specific associations; (ii) extend the analysis to morpheme-level PMI for agglutinative languages; (iii) add Arabic and African languages for broader typological coverage; and (iv) test whether information locality effects differ by dependency relation type (e.g. subject vs. object vs. modifier arcs).

References

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- [3] Gildea, D., & Jaeger, T. F. (2015). Human languages order information efficiently. *arXiv:1510.02057*.
- [4] Kahane, S., Courtin, M., & Gerdes, K. (2017). SUD: Surface Syntactic Universal Dependencies. *Proceedings of TLT16*.
- [5] Levy, R. (2008). Expectation-based syntactic comprehension. *Cognition*, 106(3), 1126–1177.
- [6] Liu, H. (2008). Dependency distance as a metric of language comprehension difficulty. *Journal of Cognitive Science*, 9(2), 159–191.

Appendix: Full Python Implementation

Full notebook (Google Colab):

https://colab.research.google.com/drive/1sf_CF5EkWUdslyIUj21wqes3GAY6H3is?usp=sharing

Git Repo:

<https://github.com/ssnehaaa/Information-Locality-and-Dependency-Distance-CGS410.git>

All code was written in Python 3.12 using: conllu 6.0.0, pandas, scipy, statsmodels, matplotlib, seaborn. Install via: `pip install conllu pandas scipy statsmodels matplotlib seaborn`

A1. Imports and Language File Configuration

```
1 import conllu, math, pandas as pd, numpy as np
2 from collections import Counter
3 from scipy.stats import spearmanr
4 import statsmodels.formula.api as smf
5 import matplotlib.pyplot as plt
6 import seaborn as sns
7 import warnings
8 warnings.filterwarnings("ignore")
9
10 # Map each language to its SUD .conllu file(s)
11 language_files = {
12     "English": ["en_gum-sud-train.conllu"],
13     "French": ["fr_gsd-sud-train_A.conllu", "fr_gsd-sud-train_B.conllu",
14              "fr_gsd-sud-train_C.conllu", "fr_gsd-sud-train_D.conllu",
15              "fr_gsd-sud-train_E.conllu"],
16     "German": ["de_gsd-sud-train.conllu"],
17     "Hindi": ["hi_hdtb-sud-train.conllu"],
18     "Japanese": ["ja_gsd-sud-train.conllu"],
19     "Persian": ["fa_seraji-sud-train.conllu"],
20     "Russian": ["ru_syntagrus-sud-train.conllu"],
21     "Turkish": ["tr_kenet-sud-train.conllu"],
22     "Chinese": ["zh_gsd-sud-train.conllu"],
23 }
24
25 typology_map = {
26     "English": "SVO", "French": "SVO", "Chinese": "SVO",
27     "German": "Mixed", "Russian": "Mixed",
28     "Hindi": "SOV", "Japanese": "SOV",
29     "Persian": "SOV", "Turkish": "SOV",
30 }
```

Listing 1: All imports and language-to-file mapping.

A2. Loading CoNLL-U Files

```
1 def load_sentences(filepaths):
2     """Load and merge one or more SUD .conllu files."""
3     all_sentences = []
4     for path in filepaths:
5         with open(path, "r", encoding="utf-8") as f:
6             all_sentences += conllu.parse(f.read())
7     print(f"Loaded {len(all_sentences):,} sentences from {filepaths}")
8     return all_sentences
```

Listing 2: Parse one or more .conllu files into sentence objects.

A3. Arc Extraction and Dependency Distance

```
1 def extract_pairs(sentences):
2     """
3     For every dependency arc in every sentence, extract:
4     (head_word, dependent_word, dependency_distance)
5     Filters: root arcs, punctuation, non-alphabetic tokens,
6             multi-word tokens.
7     """
8     pairs = []
9     for sent in sentences:
```

```

10     # Build token lookup: id -> token dict
11     tokens = {tok["id"]: tok
12               for tok in sent if isinstance(tok["id"], int)}
13     for tok in sent:
14         if not isinstance(tok["id"], int):
15             continue # skip multi-word tokens
16         head_id = tok["head"]
17         if head_id is None or head_id == 0:
18             continue # skip root
19         if head_id not in tokens:
20             continue # skip malformed arcs
21         h = tokens[head_id]["form"].lower().strip()
22         d = tok["form"].lower().strip()
23         if not h.isalpha() or not d.isalpha():
24             continue # skip punctuation/numbers
25         dd = abs(tok["id"] - head_id) # dependency distance
26         pairs.append((h, d, dd))
27     print(f"  Extracted {len(pairs):,} head-dependent pairs")
28     return pairs

```

Listing 3: Extract head-dependent pairs with dependency distances. Punctuation and root arcs are excluded.

A4. Pointwise Mutual Information

```

1 def compute_pmi(pairs, min_count=5):
2     """
3     PMI(h, d) = log2[ P(h,d) / (P(h) * P(d)) ]
4
5     P(h,d) = count(h,d) / total_arcs
6     P(h)   = count(h as head) / total_arcs
7     P(d)   = count(d as dependent) / total_arcs
8
9     Positive PMI -> pair co-occurs more than chance (strong association)
10    Negative PMI -> pair co-occurs less than chance (weak association)
11    """
12    pair_counts = Counter((h, d) for h, d, _ in pairs)
13    head_counts = Counter(h for h, d, _ in pairs)
14    dep_counts = Counter(d for h, d, _ in pairs)
15    total = sum(pair_counts.values())
16
17    pmi_scores = {}
18    for (h, d), c in pair_counts.items():
19        if c < min_count:
20            continue # filter rare pairs
21        p_hd = c / total
22        p_h = head_counts[h] / total
23        p_d = dep_counts[d] / total
24        pmi_scores[(h, d)] = math.log2(p_hd / (p_h * p_d))
25
26    print(f"  PMI computed for {len(pmi_scores):,} unique pairs"
27          f" (min_count={min_count})")
28    return pmi_scores

```

Listing 4: Compute PMI for every (head, dependent) pair. min_count=5 suppresses noisy estimates from rare pairs.

A5. Build DataFrame and Spearman Correlation

```

1 def build_dataframe(pairs, pmi_scores, language, typology):
2     """Merge pair list with PMI scores into a tidy DataFrame."""
3     rows = []
4     for h, d, dd in pairs:
5         pmi = pmi_scores.get((h, d))
6         if pmi is not None:
7             rows.append({"head": h, "dep": d,
8                         "DD": dd, "PMI": pmi,
9                         "language": language,
10                        "typology": typology})
11    df = pd.DataFrame(rows)
12    print(f"  DataFrame: {len(df):,} rows after PMI merge")
13    return df
14
15 # Main pipeline: loop over all languages
16
17 all_results, all_dfs = {}, []

```

```

18
19 for lang, filepaths in language_files.items():
20     print(f"\n{' '*45}\nProcessing: {lang}\n{' '*45}")
21
22     sentences = load_sentences(filepaths)
23     pairs = extract_pairs(sentences)
24     pmi_scores = compute_pmi(pairs, min_count=5)
25     typology = typology_map[lang]
26     df = build_dataframe(pairs, pmi_scores, lang, typology)
27
28     if len(df) > 10:
29         corr, pval = spearmanr(df["PMI"], df["DD"])
30         print(f" Spearman r = {corr:.4f}, p = {pval:.2e}")
31         all_results[lang] = {
32             "language": lang,
33             "typology": typology,
34             "spearman_r": corr,
35             "p_value": pval,
36             "n_pairs": len(df),
37         }
38         all_dfs.append(df)
39
40 print("\n\nAll languages processed.")
41 results_df = pd.DataFrame(all_results.values())
42 results_df = results_df.sort_values("spearman_r")
43 results_df["sig"] = results_df["p_value"] < 0.05
44 print(results_df[["language", "typology", "spearman_r",
45                  "p_value", "sig", "n_pairs"]].to_string(index=False))

```

Listing 5: Assemble the analysis DataFrame and compute Spearman correlation per language.

A6. Linear Mixed-Effects Model

```

1 # Combine all language DataFrames
2 combined = pd.concat(all_dfs, ignore_index=True)
3
4 # Subsample if too large (speed)
5 if len(combined) > 500_000:
6     combined = combined.sample(500_000, random_state=42)
7     print(f"Sampled to 500,000 rows for model fitting")
8
9 print(f"Total rows for LMM: {len(combined):,}")
10
11 # Fit: DD ~ PMI + Typology + (1 | language)
12 # Fixed effects: PMI, typology
13 # Random effect: language (random intercept)
14 model = smf.mixedlm(
15     "DD ~ PMI + C(typology)",
16     combined,
17     groups=combined["language"]
18 )
19 fitted = model.fit(method="lbfgs")
20 print(fitted.summary())

```

Listing 6: Fit a linear mixed-effects model with language as a random effect to confirm the PMI–distance relationship.

A7. Visualisation

```

1 colors = {"SVO": "#2196F3", "SOV": "#F44336", "Mixed": "#4CAF50"}
2
3 # Figure 1: Bar chart per-language Spearman r
4 fig, ax = plt.subplots(figsize=(12, 5))
5 ax.bar(
6     results_df["language"],
7     results_df["spearman_r"],
8     color=[colors[t] for t in results_df["typology"]],
9     edgecolor="black", linewidth=0.7
10 )
11 ax.axhline(0, color="black", linewidth=1.2, linestyle="--")
12 ax.set_title(
13     "Spearman Correlation: PMI vs Dependency Distance\n"
14     "(Negative = High-MI pairs are placed closer)",
15     fontsize=13
16 )
17 ax.set_xlabel("Language", fontsize=11)

```



```

18 ax.set_ylabel("Spearman r", fontsize=11)
19 ax.tick_params(axis="x", rotation=30)
20 from matplotlib.patches import Patch
21 ax.legend(handles=[Patch(color=c, label=t)
22                  for t, c in colors.items()],
23           title="Typology")
24 fig.tight_layout()
25 fig.savefig("correlation_by_language.png", dpi=150)
26 plt.show()
27
28 #         Figure 2: Box plot         by typological group
29 fig, ax = plt.subplots(figsize=(7, 5))
30 sns.boxplot(
31     data=results_df, x="typology", y="spearman_r",
32     order=["SVO", "SOV", "Mixed"],
33     palette=colors, ax=ax
34 )
35 ax.axhline(0, color="black", linestyle="--", linewidth=1)
36 ax.set_title("PMI-Distance Correlation by Typological Group",
37             fontsize=12)
38 ax.set_xlabel("Word Order Typology", fontsize=11)
39 ax.set_ylabel("Spearman r", fontsize=11)
40 fig.tight_layout()
41 fig.savefig("correlation_by_typology.png", dpi=150)
42 plt.show()
43
44 #         Save results
45 results_df.to_csv("correlation_results.csv", index=False)
46 combined[["language", "typology", "PMI", "DD"]]\
47     .to_csv("combined_data.csv", index=False)
48 print("Saved: correlation_results.csv, combined_data.csv")

```

Listing 7: Generate bar chart and box plot. Both figures saved as high-resolution PNG files.